Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 51.1 Moment of Inertia Numerical

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 51.1 Moment of Inertia Numerical

1 CLASS 51.1 Moment of Inertia Numerical

Study focus: CLASS 51.1 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000074 · EV0000075

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Cushion A slash forty-four.

Key details

- Determine the moments of inertia of the Z section about its centroidal x axis and centroidal y axis.
- Look at this Z section.
- The width of this member is 100, and the depth is 40, the width of this web is 20, and the height will be from the bottom most member till here it is 140.
- Then there is one more rectangular section here.
- We have been asked to.
- Find the moment of inertia this Z section about the centroidal X and centroidal y axis.
- Then where will be the centroidal x axis?
- Due to symmetry, we can say that.
- The centroidal x axis will be at a distance of 140 plus 20 divided by 2.
- That will be equal to 80 mm with respect to the bottommost member.
- Then where will be this y naught axis?
- Y naught axis will be exactly at the center, again due to symmetry, that will be equal to from ear to ear it will be 100.
- From year to ear it is 100.
- R in one more way, hundred minus half of this width, that will be ten.
- Twenty by two will be ten.
- With respect to this member also, it will be at a distance of 90.
- Once we know the centroidal x axis and centroidal y axis.

- Next, we can find moment of inertia of this Z section about the centroidal x axis and centroidal y axis.
- If you draw a vertical line passing through this, this will be the centroidal y-axis.
- Then how to find moment of inertia of this satisfaction about this axis?
- We know the moment of inertia formula for rectangles.
- Either you can divide it either by drawing a horizontal line or by drawing a vertical line.
- This is the first rectangle, and here also I'm going to draw a vertical line.
- I'll call it as 2.
- Then this is the third rectangle.
- Now moment of inertia of this z section about centroidal x axis will be equal to moment of inertia of the first rectangle about centroidal x axis plus.
- Moment of inertia of the second rectangle about the centroidal x-axis plus moment of inertia of the third rectangle about the centroidal axis.
- I am going to start with the first rectangle.
- The first rectangle is like this.
- Here, what is this width of the rectangle?
- It is not 100 because this part is included in the second rectangle.
- This is B.
- Into what is the depth of the rectangle?
- The centroidal x axis of the z section is here, x0.
- And it is at a distance of.
- This is if you we have considered this is the x-axis.
- With respect to x axis, this is at a distance of.

Example or application from the lecture

- It's given this way like.
- Let's get into the solution part.
- As I said before, the coordinates of the centroid are if you consider this bottommost member as x-axis and the leftmost member as y-axis, then the coordinates of this centroid will be 90,, 80.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 05:00–09:59

We know y bar is 80, so 80 at a distance of 80, it is there.

Key details

- Then, how to find inertia of this rectangle about the centroidal x-axis of the z section?
- We don't have a standard formula, so we have to apply the parallel axis theorem.
- To apply the parallaxis theorem, we have to.
- Look at the centroid of this rectangle.
- The center of the rectangle C is here, then what will be this distance?
- This will be 10.

- That is y bar is equal to h by 2, h is 20, 20 by 2, we will get as 10.
- Then from ear to ear distance we know that is 140.
- Whereas this is the centroidal axis of the z section.
- With respect to this rectangle, this is any other axis.
- What is I bar x formula for rectangle?
- That is bh cube by 12 plus area is B into H into distances, the distance between the centroidal axis of the rectangle and the centroidal x axis of the Z section.
- This distance we want that will be equal to 140 minus 80, we will get this distance.
- 140 minus 80, then plus 10.
- I'll write it here Bs eight into H s twenty B H cube by twelve.
- Area is B into H, that is 80 into 20 into distance square.
- Distance as I said before, that is 140 minus 80.
- That is 60, 60 plus 10, 70 square.
- Once we solve this, we are going to get the value as 7, 8, 9, 3, 3, 3, 3 mm to the board of 4.
- This is the moment of inertia of the first rectangle about the central x-axis of the z section.
- And one more, if you look at this.
- The second rectangle forget about the second rectangle for time being.
- If you look at the third rectangle, the centroid of the third rectangle will be here, and the distance from here to here will be 10.
- 10 from year to year it is 80.
- 80 minus 10 will give us 70.
- The centroidal x-axis of the z section will be equal to Ix.
- One moment of inertia the first rectangle about the centroidal z centroidal axis of the z section.
- Then we are left with only this rectangle.
- That will be equal to 140 plus 20.
- That is the height of this rectangle.
- That will be equals to 160.
- Both the centroid are coinciding with each other.
- The centroidal x axis of the z section and the centroidal axis of this second rectangle are coinciding.
- We know the standard formula that is BH cube by 12.
- Because this is 20 and h here is 140 plus 20.
- That is 160 cube divided by 12.
- And we'll get the answer as six eight two six mm to the power of four.
- Now we got the moment of inertia of the first rectangle, second rectangle and third rectangle about the centroidal x axis of the z section.
- So some just sum it up like Ix naught is equal to first piece Ix1 plus Ix2.

Example or application from the lecture

- Plus area we have to consider.
- If you consider this rectangle, where will be the centroid of the rectangle?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 10:00–14:59

Plus IX 3.

Key details

- If we add 1, then 2, plus again this, we are going to get the value of momentum inertia of the z section about the central x axis, that is 2261.
- 333 2 into mm to the power of 4.
- This is the moment of inertia of the z section about the centroidal x axis.
- Next, we need to find the moment of inertia of the same z section about its central y-axis.
- Here also we are going to apply the same concept.
- Moment of inertia of the Z section about its centroidal y axis will be equal to sum of the moment of inertia of these three rectangles about the same centroidal y axis.
- So first I'm going to start with the first so I.
- I y0 will be equal to I1.
- This is moment of inertia of the first rectangle plus I2, moment of inertia of the second rectangle plus I 3 moment of inertia of the third rectangle.
- So this is the rectangle, first rectangle, it's like this.
- Then here the width of the rectangle is 100.
- Then the height of the rectangle is 20.
- Next, where is the centroidal y-axis?
- The centroidal y axis is here.
- This is the y axis we have assumed in the figure.
- This is y axis, and this is the central y axis.
- Then the distance between these two is.
- Sorry, this is not hundred because this part is included in the second rectangle.
- And this is ninety.
- Because from year to year it is 80.
- 80 plus 10 is 90.
- That is what X bar value.
- Next, how to find inertia?
- This rectangle about this axis.
- Since we don't have a standard formula, we have to apply the parallel axis theorem.
- Where will be the centroidal y-axis of this particular rectangle?
- At a distance of x bar.
- X bar will be equal to b by 2 that is 80 by 2 will be 40.
- Inertia about any line will be equal to inertia about parallel centroidal axis plus area into distance square.
- This is the statement.
- Next, what is I bar Y?
- I bar Y formula for rectangle we know HB cube by twelve.
- Plus area is B into H.

- Distance is the distance between this centroidal y-axis of the rectangle and the y-nut axis.
- That will be equal to total 90, 90 minus 40.
- That will give us the distance between these two axes.
- Plus B is 20 and H is 80 into distances 90 minus 40, that is 50.
- We'll get this as four, eight, five.
- 3333 point three mm to the power of four.
- This is the moment of inertia of the first rectangle about the centroid and y-axis of the z section.
- Okay, now again here also we can apply the concept of symmetry because this centroidal y-axis is exactly at the same distance with respect to rectangle 1 and 3, and these two are identical, 1 and 3.

Example or application from the lecture

- Let's start with the first rectangle.
- Since the axis is vertical here, we should consider the centroidal y-axis of this particular.
- Let's start with the statement of parallel theorem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next, we are left with the second rectangle.

Key details

- Again, it's the same case now also.
- The centroid of this Z section itself is the centroid of this rectangle.
- That's the reason the centroidal y-axis of the Z section itself is the centroidal y-axis of the rectangle.
- What is h here?
- H is height of the rectangle that is 140 plus 20, that is 160 into.
- B's width of the rectangle 20 whole cube by 12.
- If we simplify we are going to get it as 10.
- 66.67 mm to the power of 4.
- This is the moment of initiative the second rectangle about the centroid and y axis.
- That is equals to moment of inertia of the first rectangle plus moment of inertia of the second rectangle plus moment of inertia of the third rectangle about the same central y axis.
- That is equals to it is 981.
- 333 3 into mm to the power of 4.
- This is the moment of inertia of the z section about the central y axis.
- If you come across any doubts, you can post your queries in the PESO forum.

Example or application from the lecture

- Consider that formula.
- try to solve the problems given in the assignment.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 51.1 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.1 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.1 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.1 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

C Revision and retrieval practice